In [1]:

```
# Install Ultralytics (latest version supporting YOLOv26)
!pip install -U ultralytics
```

```
Requirement already satisfied: ultralytics in C:\Users\User\AppData\Local\Program
s\Python\Python311\Lib\site-packages (8.4.18)
Requirement already satisfied: numpy>=1.23.0 in C:\Users\User\AppData\Local\Progr
ams\Python\Python311\Lib\site-packages (from ultralytics) (1.26.4)
Requirement already satisfied: matplotlib>=3.3.0 in C:\Users\User\AppData\Local\P
rograms\Python\Python311\Lib\site-packages (from ultralytics) (3.10.8)
Requirement already satisfied: opencv-python>=4.6.0 in C:\Users\User\AppData\Roam
ing\Python\Python311\site-packages (from ultralytics) (4.9.0.80)
Requirement already satisfied: pillow>=7.1.2 in C:\Users\User\AppData\Local\Progr
ams\Python\Python311\Lib\site-packages (from ultralytics) (10.4.0)
Requirement already satisfied: pyyaml>=5.3.1 in C:\Users\User\AppData\Local\Progr
ams\Python\Python311\Lib\site-packages (from ultralytics) (6.0.1)
Requirement already satisfied: requests>=2.23.0 in C:\Users\User\AppData\Local\Pr
ograms\Python\Python311\Lib\site-packages (from ultralytics) (2.32.5)
Requirement already satisfied: scipy>=1.4.1 in C:\Users\User\AppData\Local\Progra
ms\Python\Python311\Lib\site-packages (from ultralytics) (1.13.0)
Requirement already satisfied: torch>=1.8.0 in C:\Users\User\AppData\Local\Progra
ms\Python\Python311\Lib\site-packages (from ultralytics) (2.5.1+cu121)
Requirement already satisfied: torchvision>=0.9.0 in C:\Users\User\AppData\Local
\Programs\Python\Python311\Lib\site-packages (from ultralytics) (0.20.1+cu121)
Requirement already satisfied: psutil>=5.8.0 in C:\Users\User\AppData\Local\Progr
ams\Python\Python311\Lib\site-packages (from ultralytics) (5.9.8)
Requirement already satisfied: polars>=0.20.0 in C:\Users\User\AppData\Local\Prog
rams\Python\Python311\Lib\site-packages (from ultralytics) (1.38.1)
Requirement already satisfied: ultralytics-thop>=2.0.18 in C:\Users\User\AppData
\Local\Programs\Python\Python311\Lib\site-packages (from ultralytics) (2.0.18)
Requirement already satisfied: contourpy>=1.0.1 in C:\Users\User\AppData\Local\Pr
ograms\Python\Python311\Lib\site-packages (from matplotlib>=3.3.0->ultralytics)
(1.3.3)
Requirement already satisfied: cycler>=0.10 in C:\Users\User\AppData\Local\Progra
ms\Python\Python311\Lib\site-packages (from matplotlib>=3.3.0->ultralytics) (0.1
2.1)
Requirement already satisfied: fonttools>=4.22.0 in C:\Users\User\AppData\Local\P
rograms\Python\Python311\Lib\site-packages (from matplotlib>=3.3.0->ultralytics)
(4.61.1)
Requirement already satisfied: kiwisolver>=1.3.1 in C:\Users\User\AppData\Local\P
rograms\Python\Python311\Lib\site-packages (from matplotlib>=3.3.0->ultralytics)
(1.4.9)
Requirement already satisfied: packaging>=20.0 in C:\Users\User\AppData\Local\Pro
grams\Python\Python311\Lib\site-packages (from matplotlib>=3.3.0->ultralytics) (2
5.0)
Requirement already satisfied: pyparsing>=3 in C:\Users\User\AppData\Local\Progra
ms\Python\Python311\Lib\site-packages (from matplotlib>=3.3.0->ultralytics) (3.3.
1)
Requirement already satisfied: python-dateutil>=2.7 in C:\Users\User\AppData\Loca
l\Programs\Python\Python311\Lib\site-packages (from matplotlib>=3.3.0->ultralytic
s) (2.9.0.post0)
Requirement already satisfied: polars-runtime-32==1.38.1 in C:\Users\User\AppData
\Local\Programs\Python\Python311\Lib\site-packages (from polars>=0.20.0->ultralyt
ics) (1.38.1)
Requirement already satisfied: six>=1.5 in C:\Users\User\AppData\Local\Programs\P
ython\Python311\Lib\site-packages (from python-dateutil>=2.7->matplotlib>=3.3.0-
ultralytics) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in C:\Users\User\AppData
\Local\Programs\Python\Python311\Lib\site-packages (from requests>=2.23.0->ultral
ytics) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in C:\Users\User\AppData\Local\Progra
ms\Python\Python311\Lib\site-packages (from requests>=2.23.0->ultralytics) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in C:\Users\User\AppData\Local
\Programs\Python\Python311\Lib\site-packages (from requests>=2.23.0->ultralytics)
```

```
(2.6.2)
Requirement already satisfied: certifi>=2017.4.17 in C:\Users\User\AppData\Local
\Programs\Python\Python311\Lib\site-packages (from requests>=2.23.0->ultralytics)
(2025.11.12)
Requirement already satisfied: filelock in C:\Users\User\AppData\Local\Programs\P
ython\Python311\Lib\site-packages (from torch>=1.8.0->ultralytics) (3.20.3)
Requirement already satisfied: typing-extensions>=4.8.0 in C:\Users\User\AppData
\Local\Programs\Python\Python311\Lib\site-packages (from torch>=1.8.0->ultralytic
s) (4.15.0)
Requirement already satisfied: networkx in C:\Users\User\AppData\Local\Programs\P
ython\Python311\Lib\site-packages (from torch>=1.8.0->ultralytics) (3.6.1)
Requirement already satisfied: jinja2 in C:\Users\User\AppData\Local\Programs\Pyt
hon\Python311\Lib\site-packages (from torch>=1.8.0->ultralytics) (3.1.6)
Requirement already satisfied: fsspec in C:\Users\User\AppData\Local\Programs\Pyt
hon\Python311\Lib\site-packages (from torch>=1.8.0->ultralytics) (2026.2.0)
Requirement already satisfied: sympy==1.13.1 in C:\Users\User\AppData\Local\Progr
ams\Python\Python311\Lib\site-packages (from torch>=1.8.0->ultralytics) (1.13.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in C:\Users\User\AppData\Local
\Programs\Python\Python311\Lib\site-packages (from sympy==1.13.1->torch>=1.8.0->u
ltralytics) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in C:\Users\User\AppData\Local\Pro
grams\Python\Python311\Lib\site-packages (from jinja2->torch>=1.8.0->ultralytics)
(2.1.5)
```

In [2]:

```python
import torch
import os
import pandas as pd
from ultralytics import YOLO

print("Ultralytics Version:", YOLO.__module__)
print("CUDA Available:", torch.cuda.is_available())
print("GPU:", torch.cuda.get_device_name(0))
print("Capability:", torch.cuda.get_device_capability(0))
print("Torch CUDA:", torch.version.cuda)

device = 0 if torch.cuda.is_available() else "cpu"
print("Using device:", device)
```

```
Ultralytics Version: ultralytics.models.yolo.model
CUDA Available: True
GPU: NVIDIA GeForce RTX 5070 Laptop GPU
Capability: (12, 0)
Torch CUDA: 12.8
Using device: 0
```

In [3]:

```python
DATA_PATH = os.path.abspath("VehicleDataset/images/data.yaml")

if os.path.exists(DATA_PATH):
    print("Dataset YAML found ✅")
else:
    print("Dataset YAML NOT found ❌")
```

```
Dataset YAML found ✅
```

In [3]:

```python
model1 = YOLO("yolo26n.pt")

results1 = model1.train(
    data=DATA_PATH,
    epochs=25,
    imgsz=640,
    optimizer="AdamW",
```

```
    batch=4,
    lr0=0.01,
    device="0",
    project="/Ramil_machine_learning/yolo26_runs",
    name="model1_25ep"
)
```

```
Ultralytics 8.4.18  Python-3.11.9 torch-2.12.0.dev20260226+cu128 CUDA:0 (NVIDIA G
eForce RTX 5070 Laptop GPU, 8151MiB)
engine\trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augm
ent=randaugment, batch=4, bgr=0.0, box=7.5, cache=False, cfg=None, classes=None,
close_mosaic=10, cls=0.5, compile=False, conf=None, copy_paste=0.0, copy_paste_mo
de=flip, cos_lr=False, cutmix=0.0, data=C:\Ramil_machine_learning\VehicleDataset
\images\data.yaml, degrees=0.0, deterministic=True, device=0, dfl=1.5, dnn=False,
dropout=0.0, dynamic=False, embed=None, end2end=None, epochs=25, erasing=0.4, exi
st_ok=False, fliplr=0.5, flipud=0.0, format=torchscript, fraction=1.0, freeze=Non
e, half=False, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=640, int8=False, iou=0.7,
keras=False, kobj=1.0, line_width=None, lr0=0.01, lrf=0.01, mask_ratio=4, max_det
=300, mixup=0.0, mode=train, model=yolo26n.pt, momentum=0.937, mosaic=1.0, multi_
scale=0.0, name=model1_25ep4, nbs=64, nms=False, opset=None, optimize=False, opti
mizer=AdamW, overlap_mask=True, patience=100, perspective=0.0, plots=True, pose=1
2.0, pretrained=True, profile=False, project=/Ramil_machine_learning/yolo26_runs,
rect=False, resume=False, retina_masks=False, rle=1.0, save=True, save_conf=Fals
e, save_crop=False, save_dir=C:\Ramil_machine_learning\yolo26_runs\model1_25ep4,
save_frames=False, save_json=False, save_period=-1, save_txt=False, scale=0.5, se
ed=0, shear=0.0, show=False, show_boxes=True, show_conf=True, show_labels=True, s
implify=True, single_cls=False, source=None, split=val, stream_buffer=False, task
=detect, time=None, tracker=botsort.yaml, translate=0.1, val=True, verbose=True,
vid_stride=1, visualize=False, warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_mome
ntum=0.8, weight_decay=0.0005, workers=8, workspace=None
Overriding model.yaml nc=80 with nc=17

                   from  n    params  module
arguments
  0                  -1  1       464  ultralytics.nn.modules.conv.Conv
[3, 16, 3, 2]
  1                  -1  1      4672  ultralytics.nn.modules.conv.Conv
[16, 32, 3, 2]
  2                  -1  1      6640  ultralytics.nn.modules.block.C3k2
[32, 64, 1, False, 0.25]
  3                  -1  1     36992  ultralytics.nn.modules.conv.Conv
[64, 64, 3, 2]
  4                  -1  1     26080  ultralytics.nn.modules.block.C3k2
[64, 128, 1, False, 0.25]
  5                  -1  1    147712  ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]
  6                  -1  1     87040  ultralytics.nn.modules.block.C3k2
[128, 128, 1, True]
  7                  -1  1    295424  ultralytics.nn.modules.conv.Conv
[128, 256, 3, 2]
  8                  -1  1    346112  ultralytics.nn.modules.block.C3k2
[256, 256, 1, True]
  9                  -1  1    164608  ultralytics.nn.modules.block.SPPF
[256, 256, 5, 3, True]
 10                  -1  1    249728  ultralytics.nn.modules.block.C2PSA
[256, 256, 1]
 11                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 12             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 13                  -1  1    119808  ultralytics.nn.modules.block.C3k2
[384, 128, 1, True]
 14                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 15             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 16                  -1  1     34304  ultralytics.nn.modules.block.C3k2
```

```
[256, 64, 1, True]
 17                  -1  1     36992  ultralytics.nn.modules.conv.Conv
[64, 64, 3, 2]
 18            [-1, 13]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 19                  -1  1     95232  ultralytics.nn.modules.block.C3k2
[192, 128, 1, True]
 20                  -1  1    147712  ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]
 21            [-1, 10]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 22                  -1  1    463104  ultralytics.nn.modules.block.C3k2
[384, 256, 1, True, 0.5, True]
 23        [16, 19, 22]  1    247806  ultralytics.nn.modules.head.Detect
[17, 1, True, [64, 128, 256]]
YOLO26n summary: 260 layers, 2,510,430 parameters, 2,510,430 gradients, 5.8 GFLOP
s

Transferred 606/708 items from pretrained weights
AMP: running Automatic Mixed Precision (AMP) checks...
AMP: checks passed
train: Fast image access  (ping: 0.10.0 ms, read: 64.118.1 MB/s, size: 57.4 KB)
train: Scanning C:\Ramil_machine_learning\VehicleDataset\images\train\labels.cach
e... 3026 images, 342 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 3026/3026  0.0s
WARNING Box and segment counts should be equal, but got len(segments) = 114, len
(boxes) = 14726. To resolve this only boxes will be used and all segments will be
removed. To avoid this please supply either a detect or segment dataset, not a de
tect-segment mixed dataset.
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=
(3, 7)), ToGray(p=0.01, num_output_channels=3, method='weighted_average'), CLAHE
(p=0.01, clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))
val: Fast image access  (ping: 0.30.2 ms, read: 45.122.3 MB/s, size: 48.8 KB)
val: Scanning C:\Ramil_machine_learning\VehicleDataset\images\valid\labels.cach
e... 289 images, 28 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 289/289  0.0s
WARNING Box and segment counts should be equal, but got len(segments) = 9, len(bo
xes) = 1601. To resolve this only boxes will be used and all segments will be rem
oved. To avoid this please supply either a detect or segment dataset, not a detec
t-segment mixed dataset.
optimizer: AdamW(lr=0.01, momentum=0.937) with parameter groups 114 weight(decay=
0.0), 126 weight(decay=0.0005), 126 bias(decay=0.0)
Plotting labels to C:\Ramil_machine_learning\yolo26_runs\model1_25ep4\labels.jp
g...
Image sizes 640 train, 640 val
Using 8 dataloader workers
Logging results to C:\Ramil_machine_learning\yolo26_runs\model1_25ep4
Starting training for 25 epochs...

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       1/25      0.734G      1.865      3.158   0.009703          5        640: 10
0% ━━━━━━━━━━━━ 757/757 3.0it/s 4:12<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 2.2it/s 16.8s0.1s
                   all        289       1601      0.299      0.196      0.162
0.0795

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       2/25       0.83G      1.862      2.441   0.009826         10        640: 10
0% ━━━━━━━━━━━━ 757/757 3.7it/s 3:25<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 7.4it/s 5.0s0.1s
```

```
                   all        289       1601      0.319      0.285      0.253
0.13

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       3/25      0.83G      1.795      2.132   0.009451         28        640: 10
0% ━━━━━━━━━━━━ 757/757 4.1it/s 3:04<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 7.7it/s 4.8s0.1s
                   all        289       1601      0.563      0.355      0.323
0.168

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       4/25     0.832G      1.779      1.981   0.009275          6        640: 10
0% ━━━━━━━━━━━━ 757/757 4.1it/s 3:05<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 7.1it/s 5.2s0.1s
                   all        289       1601      0.546      0.347      0.313
0.168

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       5/25     0.832G      1.739      1.849   0.008814         12        640: 10
0% ━━━━━━━━━━━━ 757/757 4.0it/s 3:11<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 6.6it/s 5.6s0.2s
                   all        289       1601      0.562       0.41      0.373
0.209

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       6/25     0.832G      1.709      1.788   0.008645         10        640: 10
0% ━━━━━━━━━━━━ 757/757 3.8it/s 3:22<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 7.1it/s 5.2s0.2s
                   all        289       1601      0.523      0.421      0.364
0.204

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       7/25     0.832G      1.671      1.703     0.0084          4        640: 10
0% ━━━━━━━━━━━━ 757/757 4.2it/s 2:59<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 8.4it/s 4.4s0.1s
                   all        289       1601      0.674      0.409      0.406
0.233

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       8/25     0.832G      1.644      1.635   0.008189          7        640: 10
0% ━━━━━━━━━━━━ 757/757 3.8it/s 3:19<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 6.6it/s 5.6s0.2s
                   all        289       1601      0.654      0.412      0.422
0.24

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       9/25     0.832G      1.628      1.541   0.007936          8        640: 10
0% ━━━━━━━━━━━━ 757/757 3.7it/s 3:23<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 7.5it/s 4.9s0.2s
                   all        289       1601      0.667      0.436      0.435
0.237

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
```

```
      10/25      0.832G      1.601      1.477    0.00796          1        640: 10
0% ━━━━━━━━━━━━ 757/757 3.6it/s 3:29<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 7.0it/s 5.3s0.3s
                   all        289       1601      0.744      0.358      0.443
0.254

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      11/25      0.832G      1.588       1.44   0.007846         12        640: 10
0% ━━━━━━━━━━━━ 757/757 3.7it/s 3:23<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 8.1it/s 4.6s0.1s
                   all        289       1601      0.705       0.46      0.475
0.274

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      12/25      0.832G      1.557      1.388   0.007681          0        640: 10
0% ━━━━━━━━━━━━ 757/757 3.9it/s 3:13<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 7.9it/s 4.7s0.1s
                   all        289       1601      0.654      0.483      0.469
0.269

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      13/25      0.832G      1.521      1.364   0.007325         12        640: 10
0% ━━━━━━━━━━━━ 757/757 3.6it/s 3:28<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 7.9it/s 4.7s0.1s
                   all        289       1601      0.542      0.475      0.462
0.275

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      14/25      0.832G      1.531      1.316   0.007287          9        640: 10
0% ━━━━━━━━━━━━ 757/757 3.6it/s 3:28<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 7.1it/s 5.2s0.2s
                   all        289       1601      0.772      0.405      0.484
0.286

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      15/25      0.832G       1.51      1.286   0.007055         21        640: 10
0% ━━━━━━━━━━━━ 757/757 3.9it/s 3:15<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 8.4it/s 4.4s0.1s
                   all        289       1601      0.611      0.502      0.497
0.289
Closing dataloader mosaic
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=
(3, 7)), ToGray(p=0.01, num_output_channels=3, method='weighted_average'), CLAHE
(p=0.01, clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      16/25      0.832G       1.45      1.155   0.007774         10        640: 10
0% ━━━━━━━━━━━━ 757/757 3.9it/s 3:13<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 6.3it/s 5.9s0.1s
                   all        289       1601      0.661      0.475      0.512
0.318

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
```

```
      17/25      0.832G      1.456      1.118   0.007712          3        640: 10
0% ━━━━━━━━━━━━ 757/757 3.9it/s 3:13<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 7.4it/s 5.0s0.2s
                   all        289       1601      0.627       0.56      0.523
0.315

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      18/25      0.832G      1.412      1.051   0.007453          8        640: 10
0% ━━━━━━━━━━━━ 757/757 3.8it/s 3:18<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 7.9it/s 4.7s0.1s
                   all        289       1601       0.59      0.533      0.518
0.315

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      19/25      0.832G      1.395      1.005    0.00733          5        640: 10
0% ━━━━━━━━━━━━ 757/757 4.1it/s 3:04<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 7.2it/s 5.1s0.1s
                   all        289       1601      0.678      0.518      0.524
0.32

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      20/25      0.832G      1.383     0.9896   0.007088          6        640: 10
0% ━━━━━━━━━━━━ 757/757 3.9it/s 3:12<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 8.3it/s 4.5s0.1s
                   all        289       1601      0.673      0.524      0.567
0.353

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      21/25      0.832G      1.366     0.9614   0.007102          2        640: 10
0% ━━━━━━━━━━━━ 757/757 3.8it/s 3:19<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 9.4it/s 3.9s0.2s
                   all        289       1601      0.613      0.527       0.55
0.341

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      22/25      0.832G       1.34     0.9246   0.006884          7        640: 10
0% ━━━━━━━━━━━━ 757/757 3.9it/s 3:12<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 8.5it/s 4.3s0.1s
                   all        289       1601      0.698      0.517      0.565
0.349

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      23/25      0.832G      1.341     0.9193   0.006785         14        640: 10
0% ━━━━━━━━━━━━ 757/757 3.9it/s 3:14<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 8.4it/s 4.4s0.1s
                   all        289       1601      0.671      0.537      0.571
0.352

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      24/25      0.832G      1.333     0.8838   0.006856         11        640: 10
0% ━━━━━━━━━━━━ 757/757 4.0it/s 3:09<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 8.2it/s 4.5s0.1ss
```

```
                   all        289       1601        0.7      0.533      0.568
0.358

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      25/25     0.832G      1.311     0.8799   0.006715          7        640: 10
0% ━━━━━━━━━━━━ 757/757 4.1it/s 3:05<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 7.5it/s 4.9s0.2s
                   all        289       1601      0.697       0.55       0.57
0.361

25 epochs completed in 1.424 hours.
Optimizer stripped from C:\Ramil_machine_learning\yolo26_runs\model1_25ep4\weight
s\last.pt, 5.4MB
Optimizer stripped from C:\Ramil_machine_learning\yolo26_runs\model1_25ep4\weight
s\best.pt, 5.4MB

Validating C:\Ramil_machine_learning\yolo26_runs\model1_25ep4\weights\best.pt...
Ultralytics 8.4.18  Python-3.11.9 torch-2.12.0.dev20260226+cu128 CUDA:0 (NVIDIA G
eForce RTX 5070 Laptop GPU, 8151MiB)
YOLO26n summary (fused): 122 layers, 2,378,151 parameters, 0 gradients, 5.2 GFLOP
s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 37/37 11.7it/s 3.2s0.2s
                   all        289       1601      0.696       0.55       0.57
0.361
               Jeepney         36         36      0.703      0.306      0.456
0.264
                   LCV         73         82      0.889      0.878      0.917
0.538
             Trimobile          3          3          0          0     0.0505
0.0151
                   bus         52         54      0.788      0.833      0.884
0.566
                  cars          4          7          1          0     0.0574
0.0121
         container van         45         64      0.758      0.766      0.808
0.518
               e-trike         33         41      0.355       0.51      0.379
0.198
         hatchback car         19         19      0.616      0.737      0.686
0.565
                   mpv          3          3          1          0     0.0723
0.0651
              multicab          9          9       0.52      0.889      0.772
0.486
                pickup         79        117      0.785      0.747      0.802
0.462
                 sedan        167        406      0.796      0.847      0.893
0.618
                   suv        165        318      0.668      0.774      0.774
0.518
         trailer truck          6          6          1          0     0.0266
0.0133
              tricycle        134        334      0.676      0.677      0.725
0.403
                 truck         35         35      0.517      0.514      0.505
0.334
                   van         67         67      0.769      0.866      0.879
0.565
```

```
Speed: 0.3ms preprocess, 4.4ms inference, 0.0ms loss, 0.4ms postprocess per image
Results saved to C:\Ramil_machine_learning\yolo26_runs\model1_25ep4
```

In [6]:

```python
import multiprocessing
multiprocessing.set_start_method("spawn", force=True)

model2 = YOLO("yolo26s.pt")

results2 = model2.train(
    data=DATA_PATH,
    epochs=30,
    imgsz=640,
    optimizer="SGD",
    batch=20,
    lr0=0.001,
    workers =2,
    device="0",
    project="/Ramil_machine_learning/yolo26_runs",
    name="model2_30ep"
)
```

```
Ultralytics 8.4.18  Python-3.11.9 torch-2.12.0.dev20260226+cu128 CUDA:0 (NVIDIA G
eForce RTX 5070 Laptop GPU, 8151MiB)
engine\trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augm
ent=randaugment, batch=20, bgr=0.0, box=7.5, cache=False, cfg=None, classes=None,
close_mosaic=10, cls=0.5, compile=False, conf=None, copy_paste=0.0, copy_paste_mo
de=flip, cos_lr=False, cutmix=0.0, data=C:\Ramil_machine_learning\VehicleDataset
\images\data.yaml, degrees=0.0, deterministic=True, device=0, dfl=1.5, dnn=False,
dropout=0.0, dynamic=False, embed=None, end2end=None, epochs=30, erasing=0.4, exi
st_ok=False, fliplr=0.5, flipud=0.0, format=torchscript, fraction=1.0, freeze=Non
e, half=False, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=640, int8=False, iou=0.7,
keras=False, kobj=1.0, line_width=None, lr0=0.001, lrf=0.01, mask_ratio=4, max_de
t=300, mixup=0.0, mode=train, model=yolo26s.pt, momentum=0.937, mosaic=1.0, multi
_scale=0.0, name=model2_30ep, nbs=64, nms=False, opset=None, optimize=False, opti
mizer=SGD, overlap_mask=True, patience=100, perspective=0.0, plots=True, pose=12.
0, pretrained=True, profile=False, project=/Ramil_machine_learning/yolo26_runs, r
ect=False, resume=False, retina_masks=False, rle=1.0, save=True, save_conf=False,
save_crop=False, save_dir=C:\Ramil_machine_learning\yolo26_runs\model2_30ep, save
_frames=False, save_json=False, save_period=-1, save_txt=False, scale=0.5, seed=
0, shear=0.0, show=False, show_boxes=True, show_conf=True, show_labels=True, simp
lify=True, single_cls=False, source=None, split=val, stream_buffer=False, task=de
tect, time=None, tracker=botsort.yaml, translate=0.1, val=True, verbose=True, vid
_stride=1, visualize=False, warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_momentu
m=0.8, weight_decay=0.0005, workers=2, workspace=None
Overriding model.yaml nc=80 with nc=17

                   from  n    params  module
arguments
  0                  -1  1       928  ultralytics.nn.modules.conv.Conv
[3, 32, 3, 2]
  1                  -1  1     18560  ultralytics.nn.modules.conv.Conv
[32, 64, 3, 2]
  2                  -1  1     26080  ultralytics.nn.modules.block.C3k2
[64, 128, 1, False, 0.25]
  3                  -1  1    147712  ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]
  4                  -1  1    103360  ultralytics.nn.modules.block.C3k2
[128, 256, 1, False, 0.25]
  5                  -1  1    590336  ultralytics.nn.modules.conv.Conv
[256, 256, 3, 2]
  6                  -1  1    346112  ultralytics.nn.modules.block.C3k2
[256, 256, 1, True]
  7                  -1  1   1180672  ultralytics.nn.modules.conv.Conv
[256, 512, 3, 2]
  8                  -1  1   1380352  ultralytics.nn.modules.block.C3k2
[512, 512, 1, True]
  9                  -1  1    656896  ultralytics.nn.modules.block.SPPF
[512, 512, 5, 3, True]
 10                  -1  1    990976  ultralytics.nn.modules.block.C2PSA
[512, 512, 1]
 11                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 12             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 13                  -1  1    477184  ultralytics.nn.modules.block.C3k2
[768, 256, 1, True]
 14                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 15             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 16                  -1  1    136192  ultralytics.nn.modules.block.C3k2
```

```
[512, 128, 1, True]
 17                  -1  1    147712  ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]
 18            [-1, 13]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 19                  -1  1    378880  ultralytics.nn.modules.block.C3k2
[384, 256, 1, True]
 20                  -1  1    590336  ultralytics.nn.modules.conv.Conv
[256, 256, 3, 2]
 21            [-1, 10]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 22                  -1  1   1843712  ultralytics.nn.modules.block.C3k2
[768, 512, 1, True, 0.5, True]
 23        [16, 19, 22]  1    945022  ultralytics.nn.modules.head.Detect
[17, 1, True, [128, 256, 512]]
YOLO26s summary: 260 layers, 9,961,022 parameters, 9,961,022 gradients, 22.6 GFLO
Ps

Transferred 696/708 items from pretrained weights
AMP: running Automatic Mixed Precision (AMP) checks...
AMP: checks passed
train: Fast image access  (ping: 0.10.0 ms, read: 102.123.5 MB/s, size: 57.4 KB)
train: Scanning C:\Ramil_machine_learning\VehicleDataset\images\train\labels.cach
e... 3026 images, 342 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 3026/3026  0.0s
WARNING Box and segment counts should be equal, but got len(segments) = 114, len
(boxes) = 14726. To resolve this only boxes will be used and all segments will be
removed. To avoid this please supply either a detect or segment dataset, not a de
tect-segment mixed dataset.
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=
(3, 7)), ToGray(p=0.01, num_output_channels=3, method='weighted_average'), CLAHE
(p=0.01, clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))
val: Fast image access  (ping: 0.50.1 ms, read: 33.813.1 MB/s, size: 48.8 KB)
val: Scanning C:\Ramil_machine_learning\VehicleDataset\images\valid\labels.cach
e... 289 images, 28 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 289/289  0.0s
WARNING Box and segment counts should be equal, but got len(segments) = 9, len(bo
xes) = 1601. To resolve this only boxes will be used and all segments will be rem
oved. To avoid this please supply either a detect or segment dataset, not a detec
t-segment mixed dataset.
optimizer: SGD(lr=0.001, momentum=0.937) with parameter groups 114 weight(decay=
0.0), 126 weight(decay=0.00046875), 126 bias(decay=0.0)
Plotting labels to C:\Ramil_machine_learning\yolo26_runs\model2_30ep\labels.jp
g...
Image sizes 640 train, 640 val
Using 2 dataloader workers
Logging results to C:\Ramil_machine_learning\yolo26_runs\model2_30ep
Starting training for 30 epochs...

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       1/30      6.02G       1.56      5.611   0.006757         43        640: 10
0% ━━━━━━━━━━━━ 152/152 1.7it/s 1:300.4sss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 1.3s/it 10.6s.3s0
                   all        289       1601      0.149     0.0514     0.0278
0.0186

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       2/30      5.81G      1.457      3.685    0.00633         33        640: 10
0% ━━━━━━━━━━━━ 152/152 3.4it/s 45.2s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.5it/s 2.3s0.3s
```

```
                   all        289       1601      0.366      0.113     0.0818
0.0491

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       3/30      5.77G      1.441      2.672    0.00608         44        640: 10
0% ━━━━━━━━━━━━ 152/152 3.4it/s 44.9s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.4it/s 2.4s0.3s
                   all        289       1601      0.611      0.204      0.194
0.119

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       4/30      5.78G      1.415      2.008   0.005995         23        640: 10
0% ━━━━━━━━━━━━ 152/152 3.1it/s 48.4s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.1it/s 2.6s0.4s
                   all        289       1601       0.59       0.29       0.31
0.194

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       5/30      5.86G      1.399        1.7   0.005848         37        640: 10
0% ━━━━━━━━━━━━ 152/152 3.1it/s 49.8s0.4ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.0it/s 2.7s0.4s
                   all        289       1601      0.665      0.353      0.361
0.226

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       6/30      5.86G      1.371      1.524   0.005734         46        640: 10
0% ━━━━━━━━━━━━ 152/152 2.9it/s 52.9s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 2.7it/s 3.0s0.4ss
                   all        289       1601      0.649      0.377       0.39
0.247

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       7/30      5.73G      1.362      1.418   0.005647         59        640: 10
0% ━━━━━━━━━━━━ 152/152 3.0it/s 49.9s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.1it/s 2.6s0.4s
                   all        289       1601      0.626      0.448      0.443
0.278

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       8/30      5.82G      1.345      1.324   0.005555         66        640: 10
0% ━━━━━━━━━━━━ 152/152 2.9it/s 52.2s0.4ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.0it/s 2.7s0.4s
                   all        289       1601      0.722      0.459      0.461
0.293

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       9/30      5.76G      1.342      1.271   0.005592         40        640: 10
0% ━━━━━━━━━━━━ 152/152 3.0it/s 50.2s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.4it/s 2.4s0.4s
                   all        289       1601      0.627      0.473      0.494
0.318

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
```

```
      10/30       5.8G       1.33      1.215   0.005529         47        640: 10
0% ━━━━━━━━━━━━ 152/152 3.2it/s 47.9s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.1it/s 2.6s0.4s
                   all        289       1601      0.784      0.464      0.521
0.333

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      11/30      5.84G      1.329      1.172    0.00542         50        640: 10
0% ━━━━━━━━━━━━ 152/152 3.1it/s 49.6s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.2it/s 2.5s0.4s
                   all        289       1601      0.771      0.482      0.535
0.35

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      12/30      5.75G      1.317      1.129   0.005481         73        640: 10
0% ━━━━━━━━━━━━ 152/152 3.1it/s 49.2s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.2it/s 2.5s0.4s
                   all        289       1601      0.729      0.525      0.548
0.346

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      13/30       5.8G       1.31      1.099   0.005367         26        640: 10
0% ━━━━━━━━━━━━ 152/152 3.1it/s 48.3s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.4it/s 2.4s0.4s
                   all        289       1601      0.761      0.538      0.571
0.364

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      14/30      5.82G      1.305      1.076   0.005349         40        640: 10
0% ━━━━━━━━━━━━ 152/152 3.1it/s 48.5s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.3it/s 2.4s0.4s
                   all        289       1601      0.782      0.509      0.566
0.363

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      15/30      5.86G      1.305      1.045   0.005341         54        640: 10
0% ━━━━━━━━━━━━ 152/152 3.1it/s 48.9s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.1it/s 2.6s0.4s
                   all        289       1601      0.791      0.511      0.591
0.38

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      16/30      5.79G      1.291      1.023   0.005305         42        640: 10
0% ━━━━━━━━━━━━ 152/152 3.0it/s 50.0s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.3it/s 2.5s0.4s
                   all        289       1601      0.781      0.553        0.6
0.382

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      17/30      5.78G      1.294     0.9933   0.005233         61        640: 10
0% ━━━━━━━━━━━━ 152/152 3.0it/s 49.9s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.0it/s 2.7s0.4s
```

```
                   all        289       1601      0.724      0.599      0.608
0.389

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      18/30      5.72G      1.277     0.9747   0.005245         18        640: 10
0% ━━━━━━━━━━━━ 152/152 3.0it/s 49.9s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 2.9it/s 2.7s0.4s
                   all        289       1601      0.725      0.599      0.618
0.389

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      19/30      5.73G      1.283     0.9843   0.005279         30        640: 10
0% ━━━━━━━━━━━━ 152/152 3.0it/s 51.2s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.0it/s 2.7s0.4s
                   all        289       1601      0.801      0.548       0.63
0.401

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      20/30      5.86G      1.269     0.9493   0.005243         48        640: 10
0% ━━━━━━━━━━━━ 152/152 3.0it/s 50.4s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.3it/s 2.4s0.4s
                   all        289       1601      0.738      0.607      0.628
0.394
Closing dataloader mosaic
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=
(3, 7)), ToGray(p=0.01, num_output_channels=3, method='weighted_average'), CLAHE
(p=0.01, clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      21/30      5.81G      1.244     0.8516   0.005941         13        640: 10
0% ━━━━━━━━━━━━ 152/152 3.1it/s 48.9s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.3it/s 2.4s0.4s
                   all        289       1601      0.777       0.59      0.627
0.396

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      22/30      5.85G      1.234      0.819   0.005825         36        640: 10
0% ━━━━━━━━━━━━ 152/152 3.1it/s 49.5s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 2.8it/s 2.9s0.4s
                   all        289       1601       0.79      0.584      0.633
0.4

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      23/30      5.89G      1.229     0.8086   0.005706         45        640: 10
0% ━━━━━━━━━━━━ 152/152 3.0it/s 50.0s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.3it/s 2.5s0.4s
                   all        289       1601      0.754       0.61      0.635
0.404

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      24/30      5.85G      1.222     0.7958   0.005743         25        640: 10
0% ━━━━━━━━━━━━ 152/152 2.9it/s 51.8s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 2.8it/s 2.8s0.4s
```

```
                   all        289       1601      0.819      0.561      0.644
0.415

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      25/30      5.86G      1.219     0.7821   0.005651         21        640: 10
0% ━━━━━━━━━━━━ 152/152 2.9it/s 51.7s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.0it/s 2.7s0.4s
                   all        289       1601       0.81      0.576      0.643
0.412

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      26/30      5.87G       1.22     0.7819   0.005795         10        640: 10
0% ━━━━━━━━━━━━ 152/152 3.0it/s 50.8s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.4it/s 2.4s0.4s
                   all        289       1601      0.802      0.579      0.646
0.414

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      27/30      5.84G      1.213     0.7771   0.005684         19        640: 10
0% ━━━━━━━━━━━━ 152/152 3.1it/s 49.1s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.2it/s 2.5s0.4s
                   all        289       1601      0.829      0.545      0.645
0.415

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      28/30      5.87G      1.208      0.765     0.0057         34        640: 10
0% ━━━━━━━━━━━━ 152/152 3.1it/s 49.5s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.1it/s 2.6s0.4s
                   all        289       1601      0.825      0.565       0.65
0.417

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      29/30      5.85G      1.202     0.7615   0.005661         18        640: 10
0% ━━━━━━━━━━━━ 152/152 3.0it/s 49.9s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.3it/s 2.4s0.4s
                   all        289       1601      0.809      0.578      0.651
0.417

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      30/30      5.86G      1.207     0.7618   0.005632         17        640: 10
0% ━━━━━━━━━━━━ 152/152 3.1it/s 48.5s0.3ss
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 3.0it/s 2.6s0.4s
                   all        289       1601      0.819      0.568      0.651
0.421

30 epochs completed in 0.461 hours.
Optimizer stripped from C:\Ramil_machine_learning\yolo26_runs\model2_30ep\weights
\last.pt, 20.3MB
Optimizer stripped from C:\Ramil_machine_learning\yolo26_runs\model2_30ep\weights
\best.pt, 20.3MB

Validating C:\Ramil_machine_learning\yolo26_runs\model2_30ep\weights\best.pt...
Ultralytics 8.4.18  Python-3.11.9 torch-2.12.0.dev20260226+cu128 CUDA:0 (NVIDIA G
eForce RTX 5070 Laptop GPU, 8151MiB)
```

```
YOLO26s summary (fused): 122 layers, 9,471,759 parameters, 0 gradients, 20.6 GFLO
Ps
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 8/8 2.9it/s 2.8s0.4ss
                   all        289       1601       0.82      0.567      0.652
0.421
               Jeepney         36         36      0.809      0.353      0.588
0.367
                   LCV         73         82      0.931      0.817      0.906
0.509
             Trimobile          3          3      0.822      0.333      0.421
0.203
                   bus         52         54      0.854      0.944       0.94
0.605
                  cars          4          7          1          0      0.216
0.144
         container van         45         64      0.796      0.734      0.828
0.568
               e-trike         33         41       0.46      0.561      0.549
0.256
         hatchback car         19         19      0.756      0.895      0.841
0.717
                   mpv          3          3          1          0     0.0549
0.045
              multicab          9          9      0.772      0.754      0.859
0.554
                pickup         79        117       0.83      0.761      0.856
0.522
                 sedan        167        406       0.86       0.81      0.903
0.625
                   suv        165        318      0.783      0.759      0.832
0.569
         trailer truck          6          6          1          0     0.0348
0.0235
              tricycle        134        334      0.784      0.668      0.783
0.466
                 truck         35         35      0.668      0.486      0.601
0.401
                   van         67         67      0.815      0.761      0.873
0.58
Speed: 0.5ms preprocess, 3.6ms inference, 0.0ms loss, 0.3ms postprocess per image
Results saved to C:\Ramil_machine_learning\yolo26_runs\model2_30ep
```

In [4]:

```python
import multiprocessing
multiprocessing.set_start_method("spawn", force=True)

model3 = YOLO("yolo26m.pt")

results3 = model3.train(
    data=DATA_PATH,
    epochs=40,
    imgsz=640,
    optimizer="auto",
    batch=-1,   # Auto batch
    lr0=0.0001,
    workers =2,
    device="0",
    project="/Ramil_machine_learning/yolo26_runs",
    name="model3_40ep"
)
```

```
Ultralytics 8.4.18  Python-3.11.9 torch-2.12.0.dev20260226+cu128 CUDA:0 (NVIDIA G
eForce RTX 5070 Laptop GPU, 8151MiB)
engine\trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augm
ent=randaugment, batch=-1, bgr=0.0, box=7.5, cache=False, cfg=None, classes=None,
close_mosaic=10, cls=0.5, compile=False, conf=None, copy_paste=0.0, copy_paste_mo
de=flip, cos_lr=False, cutmix=0.0, data=C:\Ramil_machine_learning\VehicleDataset
\images\data.yaml, degrees=0.0, deterministic=True, device=0, dfl=1.5, dnn=False,
dropout=0.0, dynamic=False, embed=None, end2end=None, epochs=40, erasing=0.4, exi
st_ok=False, fliplr=0.5, flipud=0.0, format=torchscript, fraction=1.0, freeze=Non
e, half=False, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=640, int8=False, iou=0.7,
keras=False, kobj=1.0, line_width=None, lr0=0.0001, lrf=0.01, mask_ratio=4, max_d
et=300, mixup=0.0, mode=train, model=yolo26m.pt, momentum=0.937, mosaic=1.0, mult
i_scale=0.0, name=model3_40ep2, nbs=64, nms=False, opset=None, optimize=False, op
timizer=auto, overlap_mask=True, patience=100, perspective=0.0, plots=True, pose=
12.0, pretrained=True, profile=False, project=/Ramil_machine_learning/yolo26_run
s, rect=False, resume=False, retina_masks=False, rle=1.0, save=True, save_conf=Fa
lse, save_crop=False, save_dir=C:\Ramil_machine_learning\yolo26_runs\model3_40ep
2, save_frames=False, save_json=False, save_period=-1, save_txt=False, scale=0.5,
seed=0, shear=0.0, show=False, show_boxes=True, show_conf=True, show_labels=True,
simplify=True, single_cls=False, source=None, split=val, stream_buffer=False, tas
k=detect, time=None, tracker=botsort.yaml, translate=0.1, val=True, verbose=True,
vid_stride=1, visualize=False, warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_mome
ntum=0.8, weight_decay=0.0005, workers=2, workspace=None
Overriding model.yaml nc=80 with nc=17

                   from  n    params  module
arguments
  0                  -1  1      1856  ultralytics.nn.modules.conv.Conv
[3, 64, 3, 2]
  1                  -1  1     73984  ultralytics.nn.modules.conv.Conv
[64, 128, 3, 2]
  2                  -1  1    111872  ultralytics.nn.modules.block.C3k2
[128, 256, 1, True, 0.25]
  3                  -1  1    590336  ultralytics.nn.modules.conv.Conv
[256, 256, 3, 2]
  4                  -1  1    444928  ultralytics.nn.modules.block.C3k2
[256, 512, 1, True, 0.25]
  5                  -1  1   2360320  ultralytics.nn.modules.conv.Conv
[512, 512, 3, 2]
  6                  -1  1   1380352  ultralytics.nn.modules.block.C3k2
[512, 512, 1, True]
  7                  -1  1   2360320  ultralytics.nn.modules.conv.Conv
[512, 512, 3, 2]
  8                  -1  1   1380352  ultralytics.nn.modules.block.C3k2
[512, 512, 1, True]
  9                  -1  1    656896  ultralytics.nn.modules.block.SPPF
[512, 512, 5, 3, True]
 10                  -1  1    990976  ultralytics.nn.modules.block.C2PSA
[512, 512, 1]
 11                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 12             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 13                  -1  1   1642496  ultralytics.nn.modules.block.C3k2
[1024, 512, 1, True]
 14                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 15             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 16                  -1  1    542720  ultralytics.nn.modules.block.C3k2
```

```
[1024, 256, 1, True]
 17                  -1  1    590336  ultralytics.nn.modules.conv.Conv
[256, 256, 3, 2]
 18            [-1, 13]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 19                  -1  1   1511424  ultralytics.nn.modules.block.C3k2
[768, 512, 1, True]
 20                  -1  1   2360320  ultralytics.nn.modules.conv.Conv
[512, 512, 3, 2]
 21            [-1, 10]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 22                  -1  1   1974784  ultralytics.nn.modules.block.C3k2
[1024, 512, 1, True, 0.5, True]
 23        [16, 19, 22]  1   2824830  ultralytics.nn.modules.head.Detect
[17, 1, True, [256, 512, 512]]
YOLO26m summary: 280 layers, 21,799,102 parameters, 21,799,102 gradients, 74.9 GF
LOPs

Transferred 756/768 items from pretrained weights
AMP: running Automatic Mixed Precision (AMP) checks...
AMP: checks passed
train: Fast image access  (ping: 0.30.2 ms, read: 20.92.9 MB/s, size: 57.4 KB)
train: Scanning C:\Ramil_machine_learning\VehicleDataset\images\train\labels.cach
e... 3026 images, 342 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 3026/3026  0.0s
WARNING Box and segment counts should be equal, but got len(segments) = 114, len
(boxes) = 14726. To resolve this only boxes will be used and all segments will be
removed. To avoid this please supply either a detect or segment dataset, not a de
tect-segment mixed dataset.
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=
(3, 7)), ToGray(p=0.01, num_output_channels=3, method='weighted_average'), CLAHE
(p=0.01, clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))
AutoBatch: Computing optimal batch size for imgsz=640 at 60.0% CUDA memory utiliz
ation.
AutoBatch: CUDA:0 (NVIDIA GeForce RTX 5070 Laptop GPU) 7.96G total, 0.21G reserve
d, 0.20G allocated, 7.54G free
      Params      GFLOPs  GPU_mem (GB)  forward (ms) backward (ms)
input                  output
    21799102       74.86         1.879         230.8           nan        (1, 3,
640, 640)                    list
    21799102       149.7         4.131         185.4           nan        (2, 3,
640, 640)                    list
    21799102       299.4         7.107         156.1           nan        (4, 3,
640, 640)                    list
    21799102       598.8        12.971         145.1           nan        (8, 3,
640, 640)                    list
    21799102        1198        23.815          6154           nan       (16, 3,
640, 640)                    list
AutoBatch: Using batch-size 2 for CUDA:0 4.22G/7.96G (53%)
train: Fast image access  (ping: 0.10.0 ms, read: 44.212.7 MB/s, size: 50.1 KB)
train: Scanning C:\Ramil_machine_learning\VehicleDataset\images\train\labels.cach
e... 3026 images, 342 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 3026/3026  0.0s
WARNING Box and segment counts should be equal, but got len(segments) = 114, len
(boxes) = 14726. To resolve this only boxes will be used and all segments will be
removed. To avoid this please supply either a detect or segment dataset, not a de
tect-segment mixed dataset.
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=
(3, 7)), ToGray(p=0.01, num_output_channels=3, method='weighted_average'), CLAHE
(p=0.01, clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))
val: Fast image access  (ping: 0.10.0 ms, read: 83.954.8 MB/s, size: 49.6 KB)
val: Scanning C:\Ramil_machine_learning\VehicleDataset\images\valid\labels.cach
```

```
e... 289 images, 28 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 289/289  0.0s
WARNING Box and segment counts should be equal, but got len(segments) = 9, len(bo
xes) = 1601. To resolve this only boxes will be used and all segments will be rem
oved. To avoid this please supply either a detect or segment dataset, not a detec
t-segment mixed dataset.
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.0001' and 'momentum=0.937' and
determining best 'optimizer', 'lr0' and 'momentum' automatically...
optimizer: AdamW(lr=0.000476, momentum=0.9) with parameter groups 124 weight(deca
y=0.0), 136 weight(decay=0.0005), 136 bias(decay=0.0)
Plotting labels to C:\Ramil_machine_learning\yolo26_runs\model3_40ep2\labels.jp
g...
Image sizes 640 train, 640 val
Using 2 dataloader workers
Logging results to C:\Ramil_machine_learning\yolo26_runs\model3_40ep2
Starting training for 40 epochs...

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       1/40       2.1G      1.442      3.637   0.006944         12        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.8it/s 6:38<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 5.2it/s 14.0s<0.1s
                   all        289       1601      0.601      0.314      0.308
0.192

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       2/40       2.1G      1.438      1.904   0.007184         16        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.2it/s 7:52<0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 7.5it/s 9.7s0.2ss
                   all        289       1601      0.516      0.398      0.373
0.224

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       3/40       2.1G      1.477      1.683   0.007728         15        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.3it/s 7:41<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 8.2it/s 8.9s0.1s
                   all        289       1601      0.716      0.379      0.413
0.252

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       4/40       2.1G      1.484      1.509   0.007934         13        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.2it/s 7:53<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 7.1it/s 10.3s0.2s
                   all        289       1601      0.641      0.478      0.444
0.269

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       5/40       2.1G      1.454      1.384   0.007433         10        640: 10
0% ━━━━━━━━━━━━ 1513/1513 2.8it/s 8:56<0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 6.7it/s 10.9s0.2s
                   all        289       1601      0.727      0.416      0.461
0.28

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       6/40       2.1G      1.441      1.324   0.007465          3        640: 10
0% ━━━━━━━━━━━━ 1513/1513 2.8it/s 9:10<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mA
```

```
P50-95): 100% ━━━━━━━━━━━━ 73/73 7.7it/s 9.5s0.1ss
                   all        289       1601      0.709      0.474       0.49
0.298

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       7/40       2.1G      1.427       1.26   0.007107         27        640: 10
0% ━━━━━━━━━━━━ 1513/1513 2.9it/s 8:46<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 8.3it/s 8.8s0.2ss
                   all        289       1601      0.682        0.5      0.551
0.337

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       8/40       2.1G       1.39      1.176   0.007103         10        640: 10
0% ━━━━━━━━━━━━ 1513/1513 2.8it/s 9:06<0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 7.9it/s 9.2s0.2s
                   all        289       1601      0.853      0.433      0.567
0.351

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       9/40       2.1G      1.382      1.134   0.007146          6        640: 10
0% ━━━━━━━━━━━━ 1513/1513 2.7it/s 9:16<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 7.3it/s 10.0s0.2s
                   all        289       1601       0.79      0.527      0.584
0.354

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      10/40       2.1G      1.352      1.078   0.006899         25        640: 10
0% ━━━━━━━━━━━━ 1513/1513 2.9it/s 8:48<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 8.1it/s 9.0s0.2ss
                   all        289       1601      0.649      0.584      0.611
0.364

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      11/40       2.1G      1.342      1.028   0.006804          7        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.2it/s 7:50<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 8.0it/s 9.1s0.1s
                   all        289       1601      0.667       0.58      0.619
0.385

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      12/40       2.1G      1.318     0.9807   0.006445          8        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.5it/s 7:12<0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 8.0it/s 9.1s0.1ss
                   all        289       1601      0.791      0.571      0.622
0.392

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      13/40       2.1G      1.304     0.9822   0.006552         14        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.2it/s 7:58<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 8.2it/s 8.9s0.1s
                   all        289       1601      0.648      0.622      0.608
0.383
```

```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      14/40       2.1G      1.309     0.9637   0.006537         23        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.1it/s 8:04<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 8.6it/s 8.5s0.1s
                   all        289       1601      0.782      0.572      0.657
0.407

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      15/40       2.1G      1.297     0.9569   0.006488          9        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.6it/s 7:06<0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 8.3it/s 8.8s0.1s
                   all        289       1601      0.674      0.652      0.621
0.397

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      16/40       2.1G      1.278     0.9086   0.006254          8        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.6it/s 7:03<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 7.9it/s 9.3s0.1s
                   all        289       1601      0.645      0.662      0.659
0.405

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      17/40       2.1G      1.266     0.8626   0.006047         11        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.7it/s 6:49<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 8.4it/s 8.7s0.1ss
                   all        289       1601      0.717      0.653      0.657
0.412

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      18/40       2.1G      1.261     0.8253   0.006367          8        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.7it/s 6:52<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 8.5it/s 8.6s0.1ss
                   all        289       1601      0.734      0.663      0.721
0.463

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      19/40       2.1G      1.238     0.8254   0.005899          2        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.6it/s 6:55<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 8.9it/s 8.2s0.1ss
                   all        289       1601      0.676      0.695      0.726
0.464

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      20/40       2.1G      1.239     0.8117   0.005945         11        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.5it/s 7:08<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 8.0it/s 9.1s0.1ss
                   all        289       1601      0.711        0.7      0.704
0.44

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      21/40       2.1G      1.228      0.788   0.005994         11        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.4it/s 7:20<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
```

```
P50-95): 100% ━━━━━━━━━━━━ 73/73 7.0it/s 10.5s0.2s
                   all        289       1601      0.797      0.634      0.712
0.455

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      22/40       2.1G      1.206     0.7498   0.005779          2        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.3it/s 7:38<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 7.1it/s 10.2s0.1s
                   all        289       1601       0.72      0.635      0.715
0.45

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      23/40       2.1G      1.194      0.739   0.005647         16        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.3it/s 7:41<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 7.8it/s 9.3s0.1ss
                   all        289       1601       0.66      0.721      0.721
0.464

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      24/40       2.1G      1.181     0.7148   0.005643         17        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.1it/s 8:04<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 7.3it/s 10.0s.1s
                   all        289       1601      0.726      0.681      0.715
0.457

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      25/40       2.1G      1.161      0.722   0.005414         10        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.1it/s 8:07<0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 6.6it/s 11.1s0.1s
                   all        289       1601      0.727      0.675      0.717
0.463

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      26/40       2.1G      1.172     0.6683   0.005575          0        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.3it/s 7:34<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 7.0it/s 10.4s0.1s
                   all        289       1601      0.753      0.642      0.711
0.461

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      27/40       2.1G      1.146     0.6537   0.005321         10        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.2it/s 7:57<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 6.8it/s 10.7s0.3s
                   all        289       1601      0.706      0.736      0.728
0.472

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      28/40       2.1G      1.129     0.6526    0.00522          8        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.2it/s 7:48<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 7.4it/s 9.9s0.3ss
                   all        289       1601      0.686      0.725      0.739
0.479
```

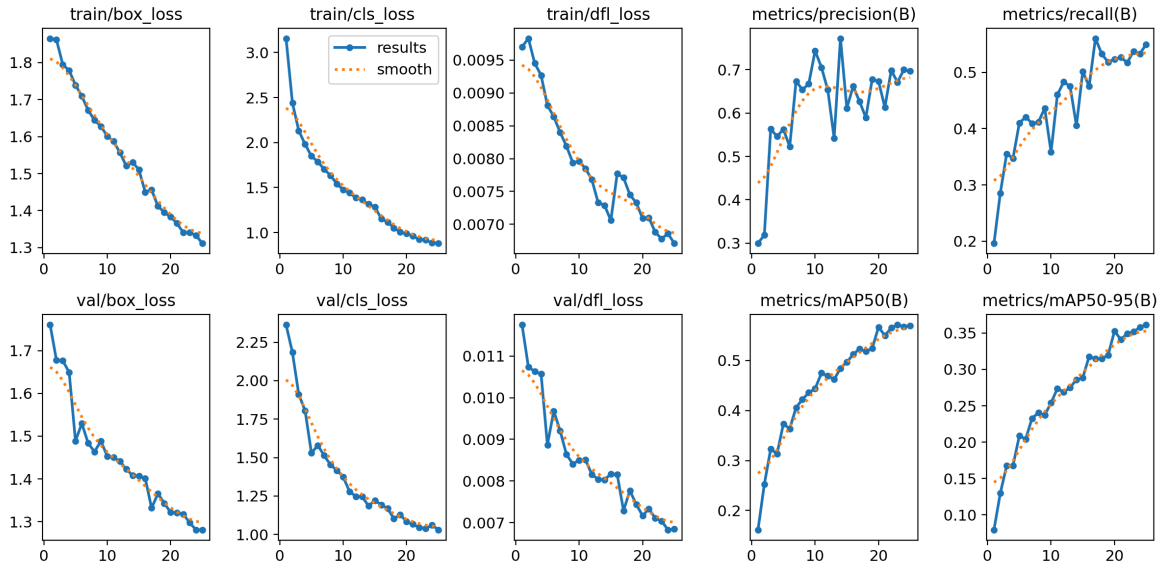
```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      29/40       2.1G      1.122     0.6279   0.005289          2        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.2it/s 7:58<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 7.7it/s 9.5s0.1s
                   all        289       1601      0.692      0.731      0.742
0.484

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      30/40       2.1G      1.114     0.6365    0.00513          4        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.2it/s 7:59<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 7.4it/s 9.9s0.2s
                   all        289       1601      0.693      0.713      0.733
0.486
Closing dataloader mosaic
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=
(3, 7)), ToGray(p=0.01, num_output_channels=3, method='weighted_average'), CLAHE
(p=0.01, clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      31/40       2.1G      1.066     0.5143   0.005393         11        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.3it/s 7:41<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 7.4it/s 9.9s0.2s
                   all        289       1601      0.663      0.741      0.727
0.482

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      32/40       2.1G      1.042     0.4921   0.005229         11        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.2it/s 7:56<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 6.7it/s 10.9s0.1s
                   all        289       1601      0.747      0.691      0.745
0.495

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      33/40       2.1G      1.033     0.4769   0.005133          9        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.2it/s 7:50<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 7.6it/s 9.6s0.1ss
                   all        289       1601       0.71      0.712       0.74
0.488

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      34/40       2.1G      1.015     0.4842   0.005079          6        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.0it/s 8:18<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 8.6it/s 8.5s0.1s
                   all        289       1601      0.715      0.728      0.743
0.491

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      35/40       2.1G      1.015       0.46   0.004984         14        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.1it/s 8:06<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 8.3it/s 8.8s0.1ss
                   all        289       1601      0.727      0.705      0.739
0.492
```

```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      36/40       2.1G     0.9973     0.4525   0.004907          5        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.2it/s 7:48<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 5.5it/s 13.2s0.2s
                   all        289       1601      0.724      0.689      0.729
0.478

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      37/40       2.1G     0.9941       0.44   0.004883          8        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.0it/s 8:30<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 7.3it/s 10.1s0.1s
                   all        289       1601       0.72      0.724      0.738
0.489

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      38/40       2.1G      0.987     0.4348   0.004764          7        640: 10
0% ━━━━━━━━━━━━ 1513/1513 3.0it/s 8:27<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 8.1it/s 9.0s0.1ss
                   all        289       1601      0.764      0.661      0.733
0.486

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      39/40       2.1G     0.9841     0.4333   0.004783          9        640: 10
0% ━━━━━━━━━━━━ 1513/1513 2.9it/s 8:42<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 7.8it/s 9.4s0.2ss
                   all        289       1601      0.702      0.724      0.731
0.485

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      40/40       2.1G     0.9797     0.4333   0.004714         15        640: 10
0% ━━━━━━━━━━━━ 1513/1513 2.8it/s 8:60<0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 6.6it/s 11.1s0.1s
                   all        289       1601      0.707      0.729      0.733
0.484

40 epochs completed in 5.432 hours.
Optimizer stripped from C:\Ramil_machine_learning\yolo26_runs\model3_40ep2\weight
s\last.pt, 44.1MB
Optimizer stripped from C:\Ramil_machine_learning\yolo26_runs\model3_40ep2\weight
s\best.pt, 44.1MB

Validating C:\Ramil_machine_learning\yolo26_runs\model3_40ep2\weights\best.pt...
Ultralytics 8.4.18  Python-3.11.9 torch-2.12.0.dev20260226+cu128 CUDA:0 (NVIDIA G
eForce RTX 5070 Laptop GPU, 8151MiB)
YOLO26m summary (fused): 132 layers, 20,362,559 parameters, 0 gradients, 67.9 GFL
OPs
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 73/73 14.0it/s 5.2s0.1s
                   all        289       1601      0.746      0.693      0.745
0.495
               Jeepney         36         36      0.954      0.578      0.791
0.501
                   LCV         73         82      0.847      0.948      0.937
0.558
             Trimobile          3          3      0.922      0.667      0.673
```

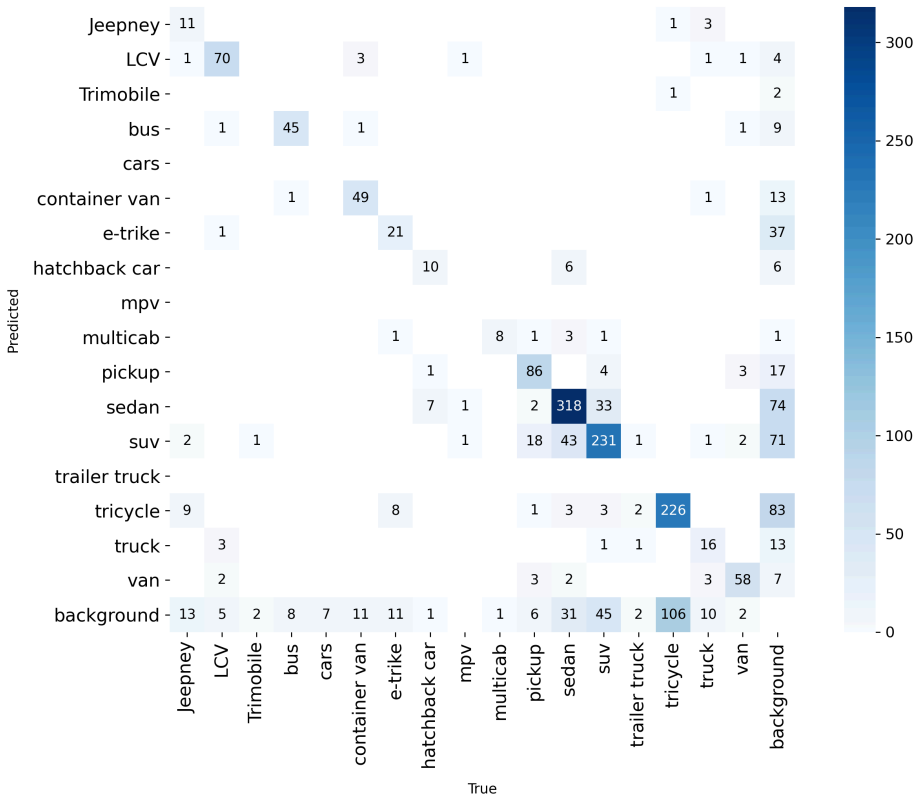
```
0.303
                   bus         52         54       0.88      0.944      0.945
0.678
                  cars          4          7       0.19      0.109     0.0655
0.0416
         container van         45         64      0.852      0.797      0.874
0.622
               e-trike         33         41       0.44      0.805      0.658
0.357
         hatchback car         19         19      0.778      0.895      0.884
0.774
                   mpv          3          3          1          0      0.592
0.489
              multicab          9          9      0.762      0.889      0.914
0.529
                pickup         79        117      0.807      0.897      0.882
0.566
                 sedan        167        406      0.869      0.862      0.919
0.63
                   suv        165        318      0.778      0.855      0.882
0.6
         trailer truck          6          6      0.182      0.167     0.0875
0.0427
              tricycle        134        334      0.781      0.754      0.824
0.505
                 truck         35         35      0.779      0.704      0.822
0.578
                   van         67         67      0.859      0.908      0.914
0.639
Speed: 0.3ms preprocess, 9.3ms inference, 0.0ms loss, 0.4ms postprocess per image
Results saved to C:\Ramil_machine_learning\yolo26_runs\model3_40ep2
```

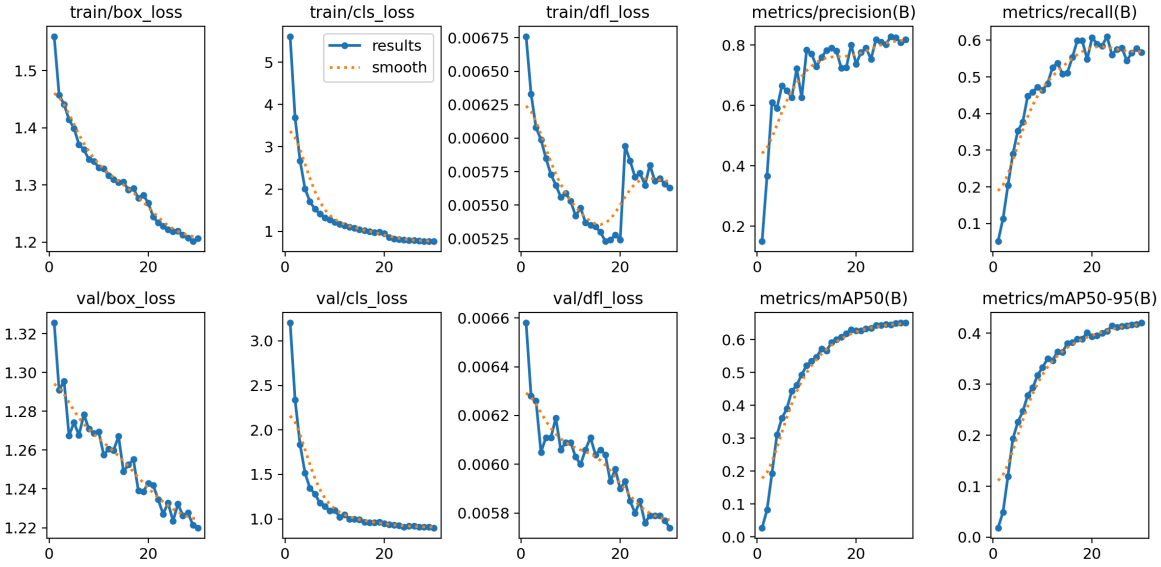
In [5]:

```python
from IPython.display import Image, display
import os

def show_results(run_name):
    base_path = f"/Ramil_machine_learning/yolo26_runs/{run_name}"

    results_img = os.path.join(base_path, "results.png")
    confusion_img = os.path.join(base_path, "confusion_matrix.png")

    if os.path.exists(results_img):
        display(Image(results_img))

    if os.path.exists(confusion_img):
        display(Image(confusion_img))

# Display all
show_results("model1_25ep4")
show_results("model2_30ep")
show_results("model3_40ep2")
```

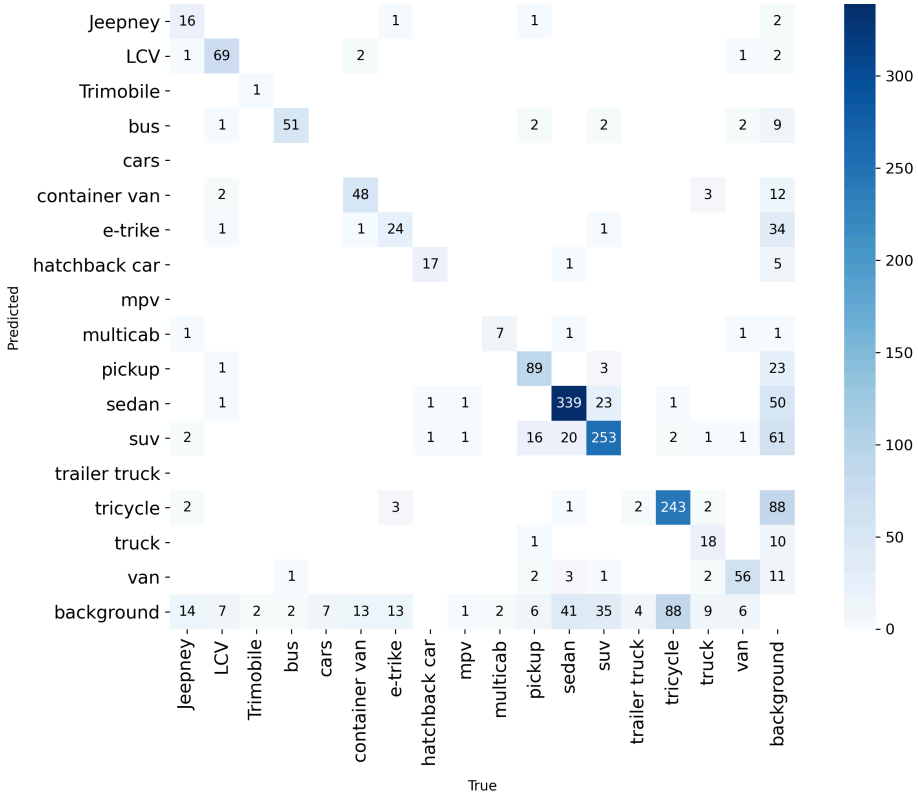


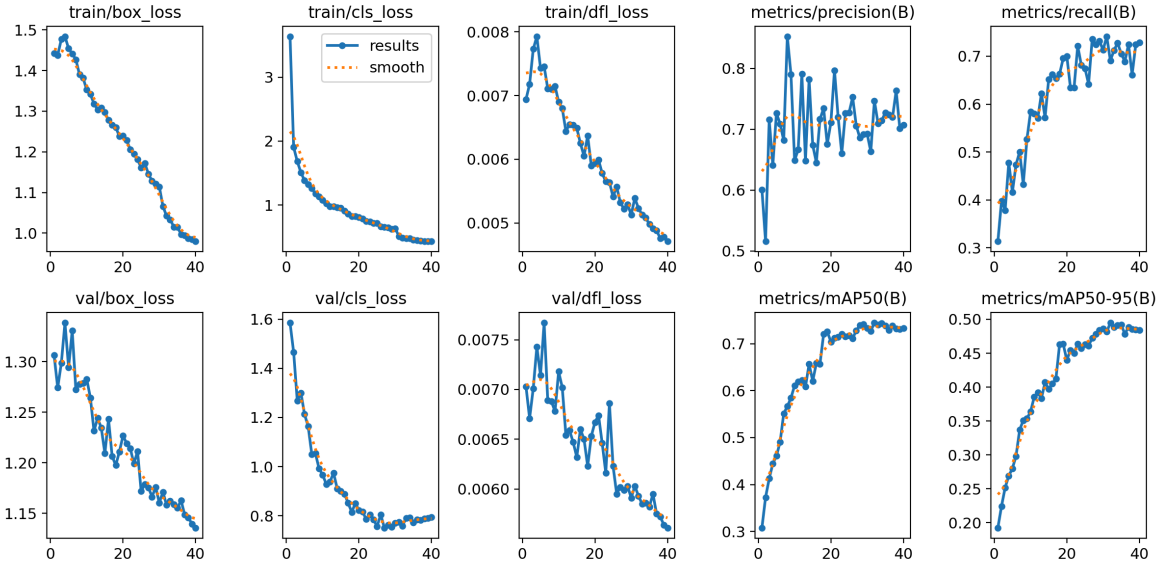
Confusion Matrix

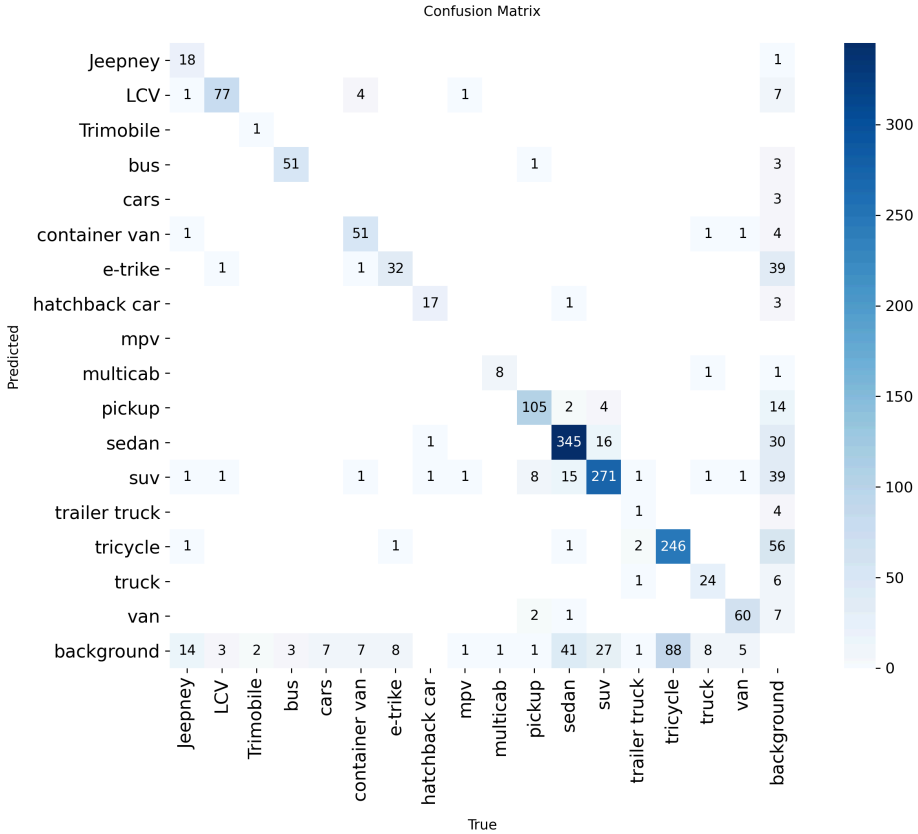




Confusion Matrix







In [6]:
```
model1_best = YOLO("/Ramil_machine_learning/yolo26_runs/model1_25ep4/weights/bes
model2_best = YOLO("/Ramil_machine_learning/yolo26_runs/model2_30ep/weights/best
model3_best = YOLO("/Ramil_machine_learning//yolo26_runs/model3_40ep2/weights/be

metrics1 = model1_best.val()
metrics2 = model2_best.val()
metrics3 = model3_best.val()
```

```
Ultralytics 8.4.18  Python-3.11.9 torch-2.12.0.dev20260226+cu128 CUDA:0 (NVIDIA G
eForce RTX 5070 Laptop GPU, 8151MiB)
YOLO26n summary (fused): 122 layers, 2,378,151 parameters, 0 gradients, 5.2 GFLOP
s
val: Fast image access  (ping: 0.10.0 ms, read: 404.1150.4 MB/s, size: 57.8 KB)
val: Scanning C:\Ramil_machine_learning\VehicleDataset\images\valid\labels.cach
e... 289 images, 28 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 289/289  0.0s
WARNING Box and segment counts should be equal, but got len(segments) = 9, len(bo
xes) = 1601. To resolve this only boxes will be used and all segments will be rem
oved. To avoid this please supply either a detect or segment dataset, not a detec
t-segment mixed dataset.
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 19/19 4.0it/s 4.8s0.2s
                   all        289       1601      0.697       0.55       0.57
0.361
               Jeepney         36         36      0.707      0.306      0.455
0.266
                   LCV         73         82      0.889      0.878      0.917
0.538
             Trimobile          3          3          0          0     0.0505
0.015
                   bus         52         54      0.789      0.833      0.884
0.566
                  cars          4          7          1          0     0.0576
0.0121
         container van         45         64      0.758      0.766      0.808
0.518
               e-trike         33         41      0.362      0.526      0.379
0.199
         hatchback car         19         19      0.617      0.737      0.686
0.564
                   mpv          3          3          1          0     0.0723
0.0651
              multicab          9          9      0.519      0.889      0.772
0.479
                pickup         79        117      0.785      0.747      0.801
0.464
                 sedan        167        406      0.794      0.845       0.89
0.617
                   suv        165        318      0.667      0.774      0.775
0.519
         trailer truck          6          6          1          0     0.0266
0.0133
              tricycle        134        334      0.672      0.677      0.724
0.403
                 truck         35         35      0.516      0.514      0.508
0.333
                   van         67         67      0.775      0.866      0.879
0.561
Speed: 4.6ms preprocess, 4.8ms inference, 0.0ms loss, 0.3ms postprocess per image
Results saved to C:\runs\detect\val9
Ultralytics 8.4.18  Python-3.11.9 torch-2.12.0.dev20260226+cu128 CUDA:0 (NVIDIA G
eForce RTX 5070 Laptop GPU, 8151MiB)
YOLO26s summary (fused): 122 layers, 9,471,759 parameters, 0 gradients, 20.6 GFLO
Ps
val: Fast image access  (ping: 0.10.0 ms, read: 488.3117.9 MB/s, size: 69.0 KB)
val: Scanning C:\Ramil_machine_learning\VehicleDataset\images\valid\labels.cach
e... 289 images, 28 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 289/289  0.0s
WARNING Box and segment counts should be equal, but got len(segments) = 9, len(bo
xes) = 1601. To resolve this only boxes will be used and all segments will be rem
```

```
oved. To avoid this please supply either a detect or segment dataset, not a detec
t-segment mixed dataset.
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 19/19 3.9it/s 4.9s0.2s
                   all        289       1601      0.821      0.566      0.651
0.42
               Jeepney         36         36      0.813      0.363      0.587
0.367
                   LCV         73         82      0.934      0.817      0.906
0.507
             Trimobile          3          3      0.823      0.333      0.421
0.203
                   bus         52         54      0.855      0.944       0.94
0.607
                  cars          4          7          1          0      0.216
0.143
         container van         45         64      0.796      0.734      0.833
0.569
               e-trike         33         41      0.462      0.561      0.549
0.257
         hatchback car         19         19      0.756      0.895      0.842
0.718
                   mpv          3          3          1          0     0.0546
0.0447
              multicab          9          9      0.772      0.753      0.859
0.532
                pickup         79        117      0.827      0.752      0.855
0.522
                 sedan        167        406      0.862       0.81      0.903
0.625
                   suv        165        318      0.784      0.754      0.833
0.569
         trailer truck          6          6          1          0     0.0347
0.0235
              tricycle        134        334      0.783      0.665      0.776
0.467
                 truck         35         35       0.67      0.486      0.594
0.396
                   van         67         67      0.813      0.761      0.873
0.585
Speed: 3.0ms preprocess, 5.9ms inference, 0.0ms loss, 0.4ms postprocess per image
Results saved to C:\runs\detect\val10
Ultralytics 8.4.18  Python-3.11.9 torch-2.12.0.dev20260226+cu128 CUDA:0 (NVIDIA G
eForce RTX 5070 Laptop GPU, 8151MiB)
YOLO26m summary (fused): 132 layers, 20,362,559 parameters, 0 gradients, 67.9 GFL
OPs
val: Fast image access  (ping: 0.00.0 ms, read: 434.5155.1 MB/s, size: 55.0 KB)
val: Scanning C:\Ramil_machine_learning\VehicleDataset\images\valid\labels.cach
e... 289 images, 28 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 289/289  0.0s
WARNING Box and segment counts should be equal, but got len(segments) = 9, len(bo
xes) = 1601. To resolve this only boxes will be used and all segments will be rem
oved. To avoid this please supply either a detect or segment dataset, not a detec
t-segment mixed dataset.
                 Class     Images  Instances      Box(P          R      mAP50  mA
P50-95): 100% ━━━━━━━━━━━━ 19/19 2.9it/s 6.7s0.4s
                   all        289       1601      0.747      0.691      0.745
0.494
               Jeepney         36         36      0.953      0.565      0.789
0.496
                   LCV         73         82      0.847      0.946      0.942
```

```
0.559
             Trimobile          3          3       0.94      0.667      0.673
0.303
                   bus         52         54      0.883      0.944      0.945
0.673
                  cars          4          7      0.154     0.0879     0.0653
0.0416
         container van         45         64      0.866      0.808      0.874
0.625
               e-trike         33         41      0.441      0.805      0.657
0.355
         hatchback car         19         19      0.782      0.895      0.883
0.774
                   mpv          3          3          1          0      0.592
0.489
              multicab          9          9      0.767      0.889      0.914
0.529
                pickup         79        117      0.812      0.897      0.881
0.564
                 sedan        167        406       0.87      0.862      0.919
0.63
                   suv        165        318      0.777      0.855      0.883
0.602
         trailer truck          6          6      0.184      0.167     0.0875
0.0427
              tricycle        134        334      0.789      0.751      0.825
0.504
                 truck         35         35      0.778      0.702      0.822
0.577
                   van         67         67      0.859      0.906      0.914
0.635
Speed: 4.0ms preprocess, 12.0ms inference, 0.0ms loss, 0.6ms postprocess per imag
e
Results saved to C:\runs\detect\val11
```

In [7]:

```python
def compute_f1(p, r):
    return 2*(p*r)/(p+r) if (p+r) != 0 else 0

performance_table = pd.DataFrame({
    "Model": ["Model1_25ep", "Model2_30ep", "Model3_40ep"],
    "mAP50": [
        metrics1.box.map50,
        metrics2.box.map50,
        metrics3.box.map50
    ],
    "Precision": [
        metrics1.box.mp,
        metrics2.box.mp,
        metrics3.box.mp
    ],
    "Recall": [
        metrics1.box.mr,
        metrics2.box.mr,
        metrics3.box.mr
    ],
    "F1 Score": [
        compute_f1(metrics1.box.mp, metrics1.box.mr),
        compute_f1(metrics2.box.mp, metrics2.box.mr),
        compute_f1(metrics3.box.mp, metrics3.box.mr)
    ]
```

```
})

performance_table
```

Out[7]:

	Model	mAP50	Precision	Recall	F1 Score
0	Model1_25ep	0.569698	0.697076	0.550403	0.615117
1	Model2_30ep	0.651495	0.820605	0.566446	0.670240
2	Model3_40ep	0.745005	0.747227	0.690995	0.718012

In [8]:

```
hyper_table = pd.DataFrame({
    "Model": ["Model1_25ep", "Model2_30ep", "Model3_40ep"],
    "Epochs": [25, 30, 40],
    "Optimizer": ["AdamW", "SGD", "Auto"],
    "Batch Size": [4, 20, "Auto"],
    "Learning Rate": [0.01, 0.001, 0.0001],
    "Image Size": [640, 640, 640]
})

hyper_table
```

Out[8]:

	Model	Epochs	Optimizer	Batch Size	Learning Rate	Image Size
0	Model1_25ep	25	AdamW	4	0.0100	640
1	Model2_30ep	30	SGD	20	0.0010	640
2	Model3_40ep	40	Auto	Auto	0.0001	640

In [ ]: